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Ex.No. 13: V2X Latency Comparison

OBJECTIVE 1:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle Speed (km/h)
```

```
speed = 20:20:120;
```

```
% Simulated End-to-End Latency (ms)
```

```
latency = [18 16 14 12 10 9];
```

```
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
```

```
disp([speed' latency'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(speed,latency,'-o','LineWidth',2)
```

```
title('Vehicle Speed vs End-to-End Latency')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

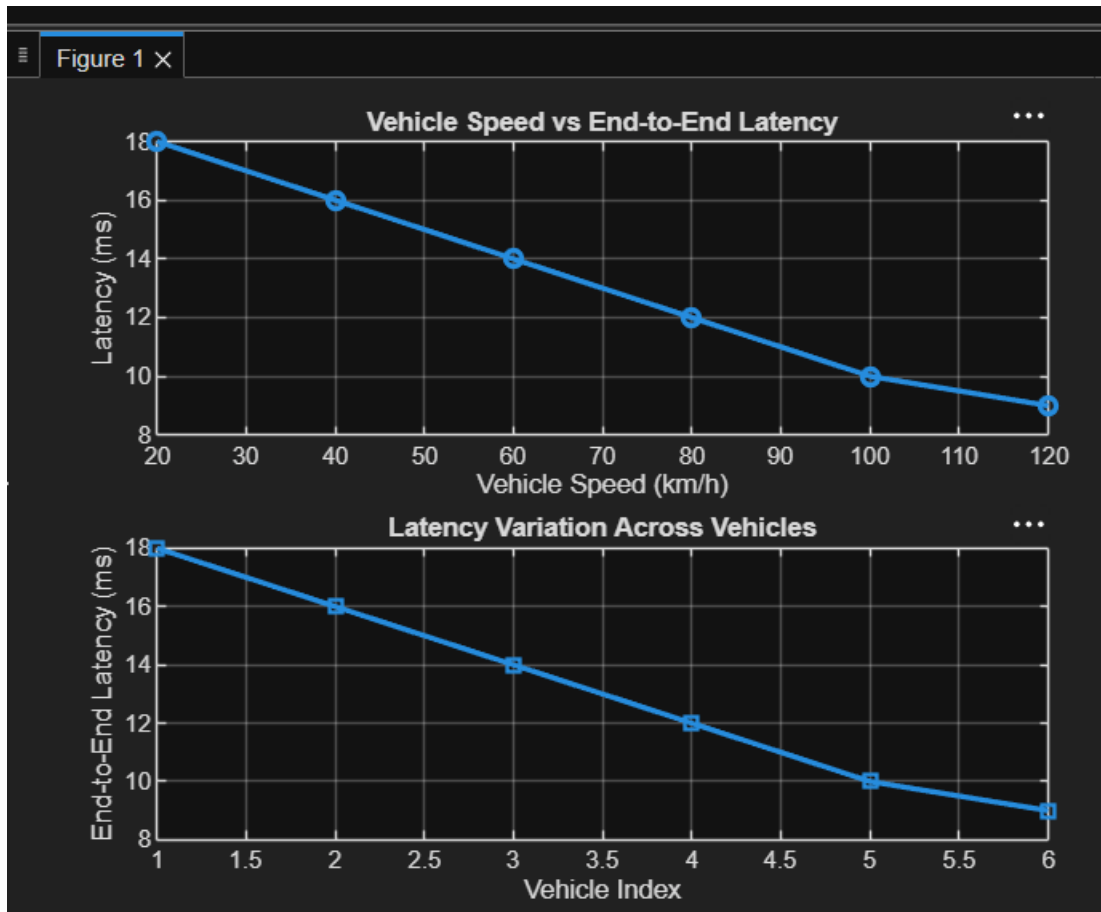
```
plot(1:length(latency),latency,'-s','LineWidth',2)
```

```
title('Latency Variation Across Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('End-to-End Latency (ms)')
```

```
grid on
```



OBJECTIVE2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle Speed (km/h)
```

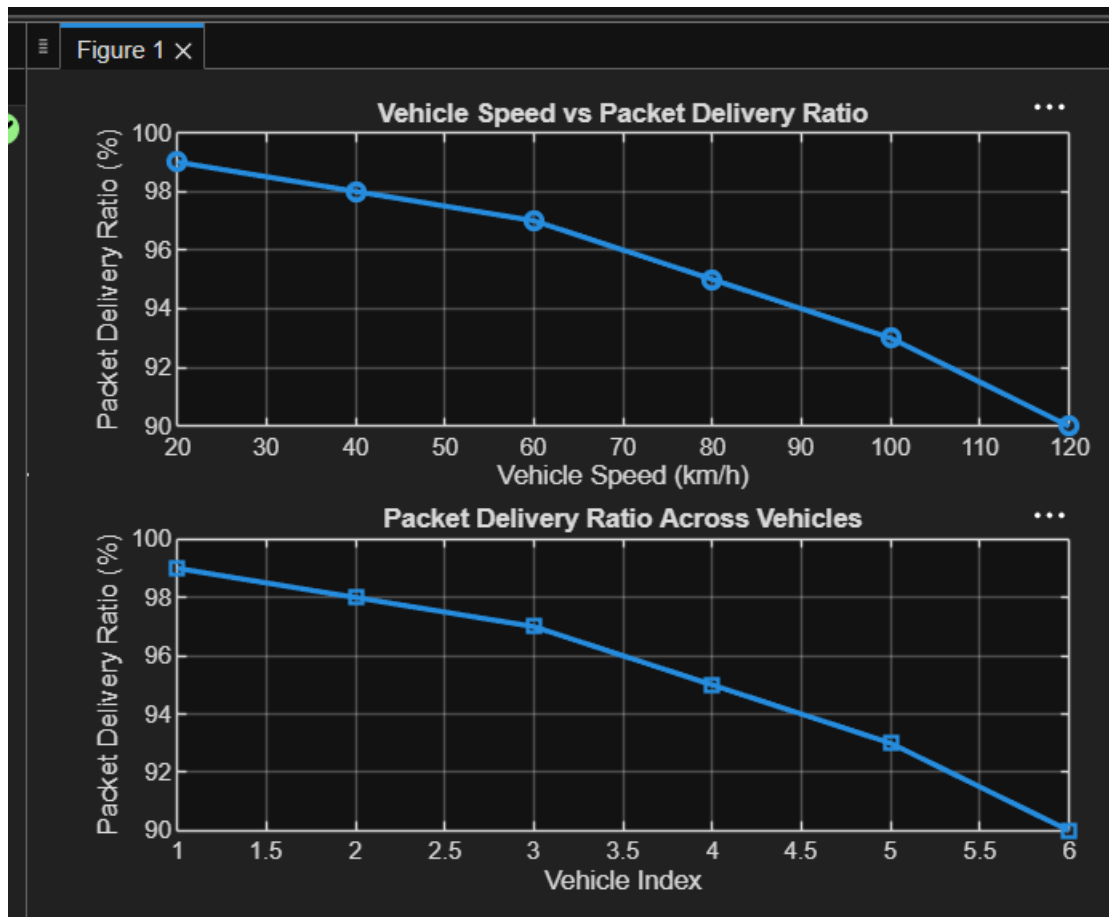
```
speed = 20:20:120;
```

```

% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];
disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')
disp([speed' PDR'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
ylabel('Packet Delivery Ratio (%)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(PDR),PDR,'-s','LineWidth',2)
title('Packet Delivery Ratio Across Vehicles')
xlabel('Vehicle Index')
ylabel('Packet Delivery Ratio (%)')
grid on

```



OBJECTIVE3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle IDs
```

```
vehicle = 1:6;
```

```
% Vehicle Speed (km/h)
```

```
speed = [30 45 60 75 90 105];
```

```
% Communication Range (m)
```

```
range = [280 260 240 220 200 180];
```

```
% Display values
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
plot(vehicle, speed, '-o', 'LineWidth', 2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(speed, range, '-s', 'LineWidth', 2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on
```

